

Literature status: quantitative convergence in the inconsistent proximal-splitting case

Automated mathematics run `fa_banach_001`, lane 2

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Status: literature already answered for open problem 1, in the form of quantitative sufficient conditions.

Source question

Florian Lauster and D. Russell Luke, *Convergence of Proximal Splitting Algorithms in $\text{CAT}(\kappa)$ Spaces and Beyond*, arXiv:2105.05691, Section 6, physical PDF page 17. Its first open direction is to determine requirements for quantitative convergence of proximal splitting methods when the individual proximal mappings do not have common fixed points—the “inconsistent case.”

Supporting answer

D. Russell Luke and Mahshid Mirhashemi, *Quantitative Convergence of Proximal Splitting Iterations in Uniformly Convex Metric Spaces*, arXiv:2605.03484. The abstract says explicitly that the theory does not assume common minima or vanishing proximal parameters. The introduction (physical pages 1–2) cites the source as reference [29], describes [29, Theorem 25] as a consistent-case convergence-rate result, and identifies quantitative convergence for inconsistent instances as the missing case addressed by the new paper.

Theorem 3.3 (physical page 9) treats cyclic and relaxed compositions

$$T = \text{prox}_{f_m, \lambda_m}^p \circ \cdots \circ \text{prox}_{f_1, \lambda_1}^p$$

and their geodesically relaxed analogues on p -uniformly convex spaces with curvature bounded above. Under its explicit regularity, fixed-point existence/invariance, and stability-gauge hypotheses, parts (iv)–(v) give quantitative convergence and, for a linear gauge, R -linear convergence. No intersection $\bigcap_j \arg \min f_j$ is assumed. Corollary 3.4, beginning on physical page 10, specializes these conclusions to Hadamard spaces and removes the general composition assumptions in the parameter range stated there.

Provenance and scope

This identification is author-aware, not merely inferred: Luke coauthored both papers, and the supporting paper explicitly cites and contrasts the source theorem before announcing the removal of common minima. It therefore belongs in `literature_already_answered`.

The answer is scoped. It supplies sufficient requirements, but not an intrinsic necessary-and-sufficient characterization. Existence/invariance and a metric-subregularity-type stability gauge

remain assumptions; in positive curvature, additional composition/relaxation hypotheses remain. The supporting paper’s conclusion leaves removal of some of these assumptions open. The source’s second open direction—automatic metric subregularity from nonlinear structural hypotheses—is not resolved and is reiterated as future work.

Evidence

The packet contains the original arXiv PDF as `source_paper.pdf` and the decisive later PDF as `supporting_paper_2605.03484.pdf`. The source question is on physical page 17; the supporting identification is on pages 1–2, with the main theorem and Hadamard corollary on pages 9–10.

References

- [1] F. Lauster and D. R. Luke, *Convergence of Proximal Splitting Algorithms in $\text{CAT}(\kappa)$ Spaces and Beyond*, arXiv:2105.05691 (2021).
- [2] D. R. Luke and M. Mirhashemi, *Quantitative Convergence of Proximal Splitting Iterations in Uniformly Convex Metric Spaces*, arXiv:2605.03484 (2026).